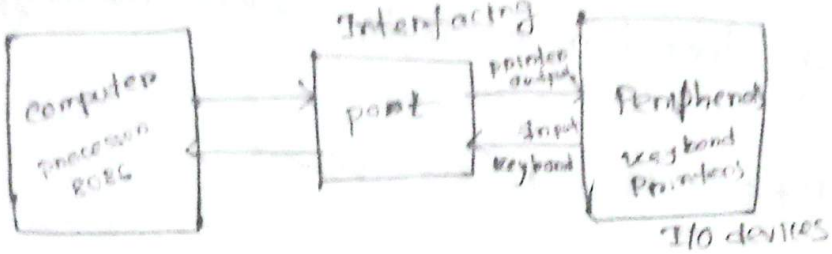
07.05.2026

1st Class

CSE 4141: Microcontroller Interfacing

- PPI - Programmable Peripheral Interfacing
- HPI - Hardware Peripheral Interfacing



GrPA
GrPB
GrPC

8085 → Ramesh Gaonkar

11.05.2026

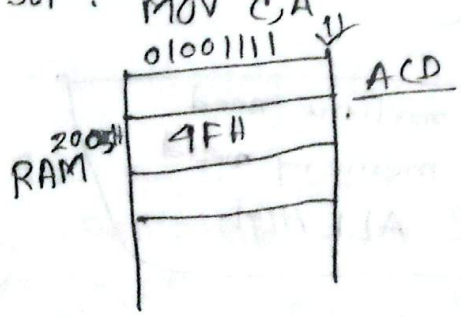
Chapter-3

- Control And status Signals

3.12 Microprocessor Communication and Bus timing

Example: The instruction code 01001111 (4FH-MOV C,A) is stored in memory location 2005H. Illustrate the data flows and list the sequence of events when the instruction code is fetched by the MPU

Soln:



- Fetch

⇒ IO/M = 1
⇒ RD = 0 → MEMR
⇒ WR

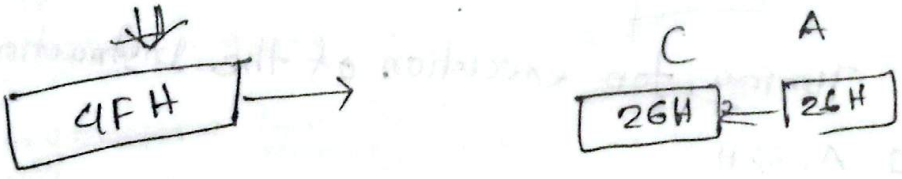
PC/IP

- Opcode

5 Instruction phase cycle stages
1. IF
2. ID
3. OF
4. EX
5. SB

Instruction phases cycle/ stages

ACD



stage-4

Exam
• How mpu communicate

Program counter → Address generation
Stack pointer → Address holden
fig 3.2 Data flow from Memory to the MPU
Exam fig 3.3 . Timing transfer of Byte from Memory to M

$$T = \frac{1}{f}$$

$f = 2\,MHz \quad T = \frac{1}{2}\,\mu s = 0.5\,\mu s$

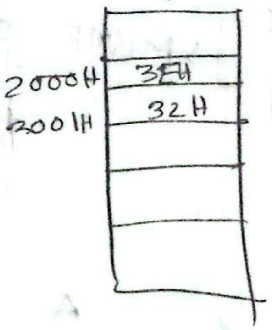
$A_{15} \cdots A_8\ AD_7 \cdots A_1\ D_1\ A_0$ ALE = 1 ↳ $A_7 \cdots A_1 A_0$

fetching এর সময়ে ALE High হবে

14.05.2026

fig. 3.5
fig: 3.6
3.21 The 8085
3.22 Opcode fetch machine cycle
3.23 Memory Read Machine cycle (Operands - Data/Address)

MOV
MVI → Accumulator register intruction
↳ MVI A,
↳ 3EH
opcode



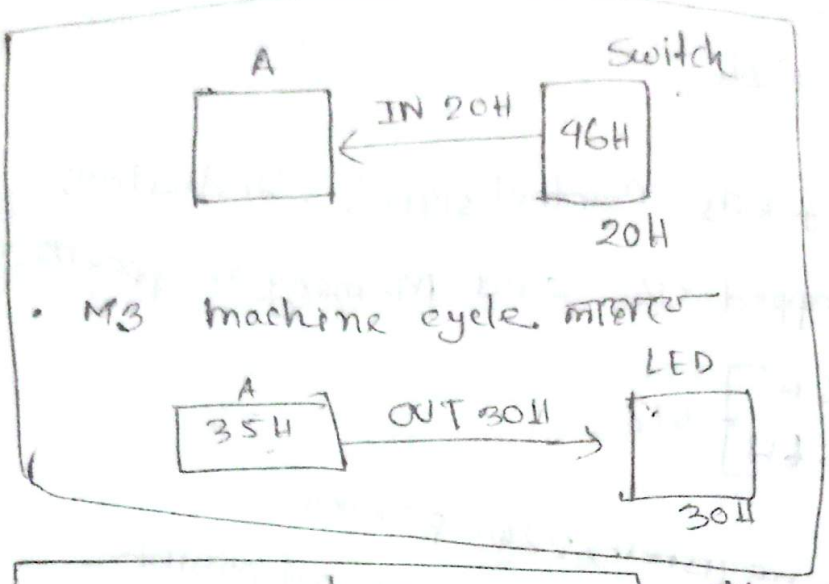
2 machine need
2 memory read
2 ALE High

fig: 3.10 8085 Timing for execution of the Instruction M
MVI A, 32H

- IN 8bit Port Address
- OUT " "

→ Exam

IN 20H → DBH
OUT 30H → D3H



$S_7 \; - \; - \; - \; S_2 S_1 S_0$
01000110

- M3 machine cycle. minimum

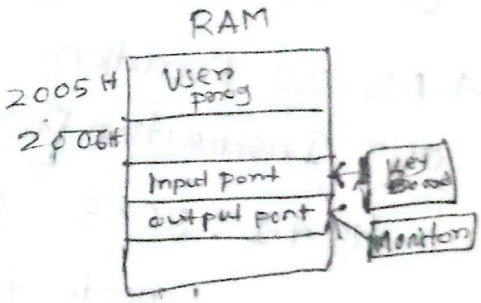
Opcode Fetch
Memory read
I/O read
→ Input

opcode Fetch
Memory read
I/O write
→ out

I/O map interfacing

→ mem. read
→ LDA 8000 H
→ STA 7000H
↳ mem write
→ 4byte

memory map interfetching



3.3 Memory Interfacing

fig 3.11

18.05.2026

Chapter 4

- Interfacing I/O Devices

4.1. Basic Interfacing concepts

- peripheral I/O Instructions

Exam
- Def.n (Intel processor)
- I/O mapped I/O
- Memory mapped I/O
  ↳ Motrola 68000

- No of Bits
  - 8bits ports
  - 16 bits ports memory address

Control signals
- IOR IOW
- MEMR, MEMW

instructions
IN, OUT
LDA, STA

- Isolated on separate ports for interfacing
  74LS373, 74LS244
  (Latch) (Buffer)

Buffer Latch: IN 20H
Buffer Latch: OUT 20H
} possible, differentiate two same address using control signals

LDA 8005H
STA 8005H
} not possible.

Exam

differentiate using No. of bits, Control signals, Instructions

Memory between I/O mapped I/O and Memory mapped I/O

IN 20H, IN = 00H → FFH - 256H
OUT 20H, OUT = 00H → FFH - 256H
} 512

LDA 8000 H
STA 8000 H
} read/write register, 2 not possible
interfacing read/write register, 2 not possible

- Out Instruction, ~~memory~~ eg/3 machine cycle need for 2 Instruction

4.1.2 I/O Execution

OUT Instruction Exam

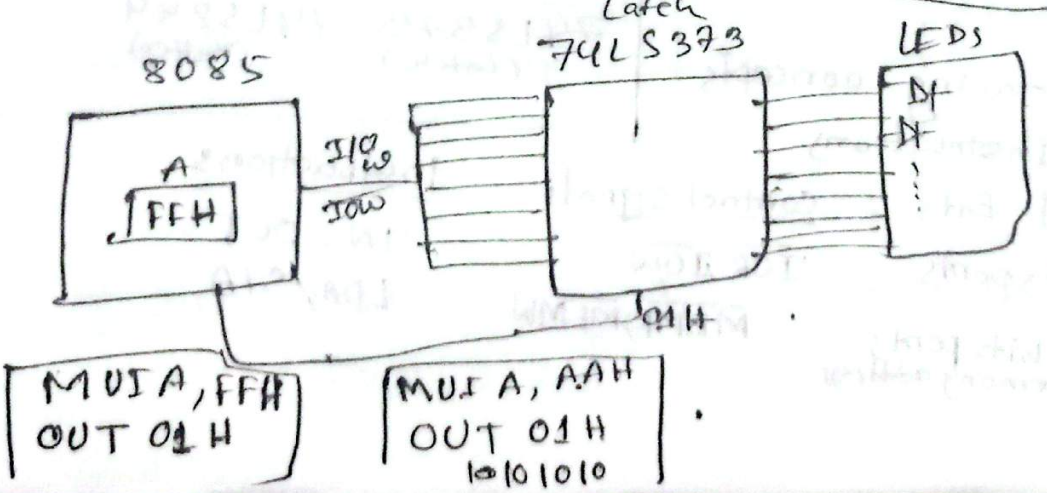
fig 4.1 : 8085 Timing for Execution of OUT instruction

1. opcode fetch
2. Operand fetch
3. execution

Instruction OUT

M_1 (opcode fetch)
M_2 (Memory read)
M_3 (I/O ~~Memory~~ write)

T4 use for decoding



OUT Instruction

- 1. IF → MEMR
2. ID
- 3. OF → MEMR
- 4. EX → I/OW
5. SB → MEMW

Input Instruction

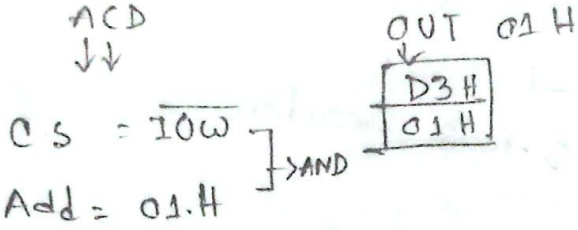
1. M_1 (opcode fetch
2. M_2 (Memory re
3. M_3 (I/O read)

- LDA 8000H
  STA 8005H
  32H 50H
  80H

MOV C, A = 4FH
MVI A, 05H

14.06.2026

Exam
5.1.3 Device Selection and data Transfer
fig: 4.3 Block Diagram of I/O Interface
fig: 4.4 Decode Logic for LED Output Port



Exam
Q. How a device select pulse generate base priority signal data
Q. কিভাবে device select pulse generate করা হয় ?

Exam
4.14. Absolute Vs. Partial Decoding → fig 4.4, fig 4.5 Solution

fig 4.4

small system use partial decoding
Large system use absolute decoding

fig: 4.5 Partial Decoding: Output Latch with Multiple Address

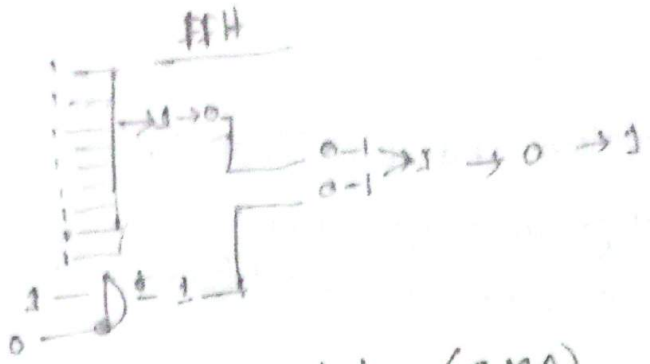
4.15

15.06.2026

4.15. Input Interfacing

fig 4.6. Decode logic for a Dip-Switch Input port (switch off)

1. Control signal IOR is used inplace of $\overline{IOW}$.



Compliment Accumulator (CMA)

fig 4.7 ~~Decode logic for a Dip-Swi~~ Address decoding. Using a 3-to-8 Decoder

	A_7 A_3 A_2 A_1 A_0		
~~FFFF~~	11111	0 0 0	F8H → output
	11111	0 1 0	FAH → Input

9.2 Interfacing

fig 4.10: seven-segment LED

fig 4.11. Interfacing DIP Switches

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fig: 9.9 Seven-segment LED: LED Segments
(a) Common-Anode LED
(b) Common-Cathode LED

fig: 4.10

fig:10

A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
1	1	1	1	0	1	0	1
F				5			



```
MVI A, 78H
OUT F5H
HLT
```

4.4 Memory-Mapped I/O:

16 bit address
· instructions LDA / STA

fig 4.12 Timing for Execution of the instruction: STA 8000H

Exam

4.42. Illustration: Safety control system using Memory-Mapped I/O technique

fig: 4.13 Memory-Mapped I/O Interfacing

```
READ: LDA FFF9H
      ~~STA~~
      CMA
      STA FFF8H
      JMP READ
```

22.06.2026

Chapter-14

PPI (Preggnamable Peripheral device)

8155 - Special Purpose PPI ⇒ 8085
Multi-Purpose

I/O Timer

8255 - General Purpose PPI ⇒ 8086 series
Single Purpose ⇒
I/O operations

8254 ⇒ Counter

8279, 8237 (PPI) ⇒ Keyboard Interfacing

fig: 14.2 Making 74LS245 Programmable

245 ⇒ 244 + 373
T R R T

T: Transmitter
R: Receiver
TR: Transceiver

fig: 14.3 A Hypothetical Programmable Device with a status Register

fig: 14.5 8155 Pin Configuration and Block Diagram

fig: 14.6 Expanded Block diagram of the 8155 (a) and Its I/O Address : Selection (b)

fig: 14.7 Interfacing 8155 I/O Ports (schemantic from the SDK-85 system)

control World (CW)

fig: 14.8 Control world Defination in the 8155 and Table of Port C Bit Assignments (b)

fig: 14.9 Interfacing 8155 I/O Ports with seven segment LE

29.06.2026

Chapter-15 Exam

- General Purpose Programmable Peripheral Devices

15.1. The 8255A Programmable peripheral interface

fig: 15.1 8255A I/O ports and their Modes

fig 14.4 Handshake signals Exam
- what is handshake signals
- why it is use?

fig 14.8

• Timer Exam

↳ counter

16 bit Down Counter

8155

8254 counter $M_2M_1T_{13} \ldots T_3T_2T_1$

$C_{13} \ldots C_LC_1C_0$

CR/SR = 20H

Port A = 21H

Port B = 22H

Port C = 23H

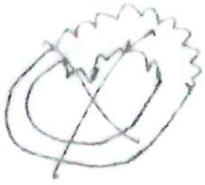
Timer LSB = 24H

Timer MSB = 25H

M = Mode

T = Timer

	M_2	M_1
Mode 0	0	0
Mode 1	0	1
Mode 2	1	0
Mode 3	1	1



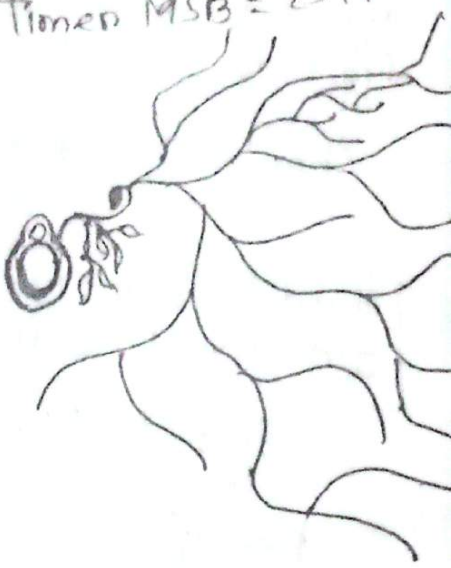


fig 14.10

Timer loading format and Modes

14.2.4 Illustration H.W

• Timer রেজিস্টার I/O ডাটাবাস control world ~~অপারেশন~~ ডাটা ফরম্যাট mode রেজিস্টার ২টি timer ও I/O ~~ও~~ control world " " অপারেশন " " " " "

02.07.2025

15.1. 8255A

• BSR Mode I/O Mode

The functions

figure 15.1 - 8255A I/O Ports and their Modes

15.1.1 Block Diagram of the 8255A

fig 15.2 of the control logic and I/O ports

Exam

- fig 15.3 8255A Chip select logic and I/O Port Addresses
- fig 15.4 8255A control World for I/O Mode

09.07.2026

fig: 15.1 : 8255A I/O ports and Their Modes (6)
fig. 15.4 8255A Control world for I/O mode

Handshake signal ছাড়া কাজের জন্য C ports use করা হয়

Handshaking : I/O an I/O control approach to synchronize I/O devices with the microprocessor.

fig: 15.5 . Interfacing 8255A I/O Ports in Mode 0

Example 15.1

A_{15} ---- $A_2 A_1 A_0$		
1000000000000 00	= 8000 H	= PA
1000000000000 01	= 8001 H	= PB
1000000000000 10	= 8002 H	= PC
1000000000000 11	= 8003 H	= CR/SR

(1000000000000 = fixed)

- control world
- program

ANI 0F H ⇒ Masking

```
PC7 ---- PC2 PC1 PC0
A = 1011 0110
M = 0000 1111
A = 0000 0110
```

ANA B
ORA C

RLC → intensive
RCL → through carry → extensive

ORI 8bit data
ANI 8bit data
ANA Reg
ORA Reg

Chapter 6

34

12.07.2026

15.13. BSR (set/Reset) Mode

- BSR Control word
- Example 15.2
- fig: 15.6 · 8255A Control Word format in the BSR Mode

control word bit 7 PC7 PC6 PC5 PC4
set/reset set reset

- Port Address

Exam

15.9. The 8254 (8253) Programmable Interval timer (H.W)

19.4.1 Block diagram of the 8254

fig: 15.23 8254

16.07.2026

Chapter-6 Introduction to 8085 Instructions

- Example 6.2
- fig. 6.5. Reading Data at Input Port and sending Data to Output Port
- Example 6.8
- fig 6.8 Input Port to Control Applicatiances
- 6.35 Illustrative Program: ORing Data from Two Input Ports
- fig 6.9: Two Input Ports to Control Output Devices

Exam 6.52 Illustrative Program: Microprocesson - Controlled Manufacturing Process

Exam fig 6.12 Input/Output ports to control Manufacturing Processes

- Problem fig 6.13. Flowchart and Program for Controlling Manufacturing Processes

program answer

S_7 . . . S_1 S_0
1 0 0
S'_7 . . . S_1 S_0
1 0 0

P_1	P_2	
S_7	S'_7	
0	0	= 00 H
0	1	= 00 H
1	0	= 00 H
1	1	= 80 H

23.07.2026

chapter-8

- Counter and time Delays

8.1. Counter and Time Delays

Counter

- Time Delay

8.11 Time Delay using One Register

- Technique to create time delay:
  1. Using single (one) Register
  2. Using Register Pair (Rp) Exam
  3. Using Loop inside loop (nested loop) Exam N.V.J

FF FF
B C

A	Flags
B	C
D	E
H	L
SP	
PC	

MVI R, 8-bit data

LXI Rp, 16-bit data

8.12 Time Delay Using a Register Pair

8.13 Time Delay Using a Loop within a Loop technique

8-3 Illustrative program: zero-to-nine (Modulo ten) counter

- Problem statement
- Delay Calculations
- fig 8.7. Program and Flowchart for a zero-to-Nine counter

9.21. Illustrative Program: Traffic Signal Controller

- Problem fig 9.13

26.07.2026

2.2.2 Memory Map

- RM0350

stm32f446re.pdf

4. CRC calculation unit

13. ADC
14. DAC
15. watchdog
22. RTC
24. I2C Interface
26. SPI, I2S, USART

NUCLEO-F446RE
GPIOC Pin 13
GPIOA Pin 5

- basic remembering
I2C1

- pullup
- pull down

#define GPIOB_AFRH

USART, I2C, SPI on TIM

- CRC, timer, NVIC, Bus Exam

27.07.2026

29.07.2026

pull up, pull down

CamScanner